Resilience Learning Activity

Data Analytics Training - Level 3

Session 5

Inferential Statistical Analysis

Learning objectives

OBJECTIVES: By the end of this session, participant should be able to:

- Understand the different aspects of inferential statistics that is, correlation, t-test and ANOVA
- Carry out computation of the statistics
- Apply inferential statistics in the context of resilience data

What is inferential statistics?

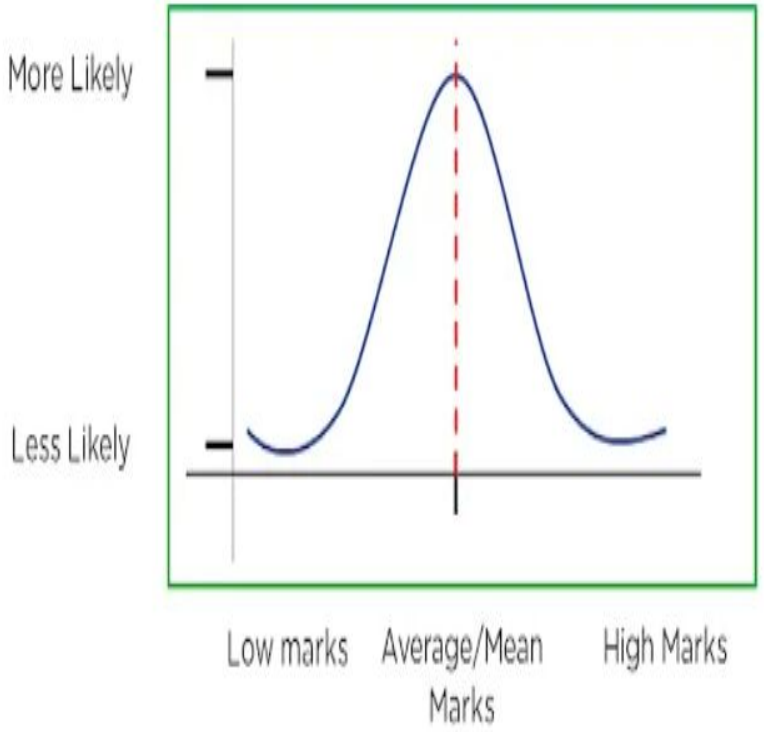
- The goal of inferential statistics is to discover some property or general pattern about a large group by studying a smaller group of people in the hopes that the results will generalize to the larger group.

- The following types of inferential statistics are presented:
  - Skewness measurement
  - Correlation.
  - Two sample t test
  - ANOVA

SKEWNESS MEASUREMENT

Normal Distribution

- A normal distribution is a continuous probability distribution for a random variable.
- A random variable is a variable whose value depends on the outcome of a random event.
- For example, flipping a coin will give you either heads or tails at random. You cannot determine with absolute certainty if the subsequent outcome is a head or a tail.
- A normal distribution or continuous probability distribution curve is bell-shaped. The peak of the curve is at the mean and the data is symmetrically distributed on either side of it.
- The mean, median, and mode are equal to each other or lie close to each other.
- Skewness is zero for a normal distribution curve.



Normal Distribution Curve

Skewness

- **Skewness** describes how much statistical data distribution is asymmetrical from the normal distribution, where distribution is equally divided on each side.

- If a distribution is not symmetrical or Normal, then it is skewed
- Two types of skewness:
    - Positively Skewed
    - Negatively Skewed

Positively Skewed

- When distribution is Positively Skewed, most of the frequency of distribution lies on the right side of distribution.
- The statistical values are bent towards the left side of the tail.
- Distribution: Mean > Median > Mode.
- The values are always positive.

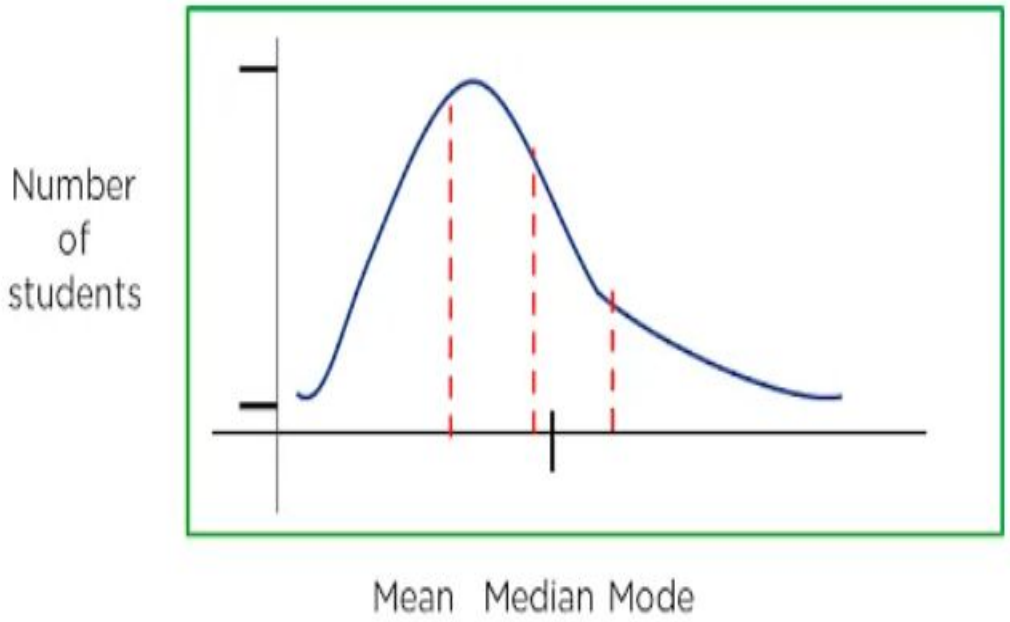


Figure: Positively Skewed

Negatively Skewed

- In a negatively skewed distribution, most of the frequency of distribution lies on the left side of distribution.

- The statistical values bend towards the right side of the tail.

- In this distribution,
  Mean < Median < Mode

- Hence these values are always negative.

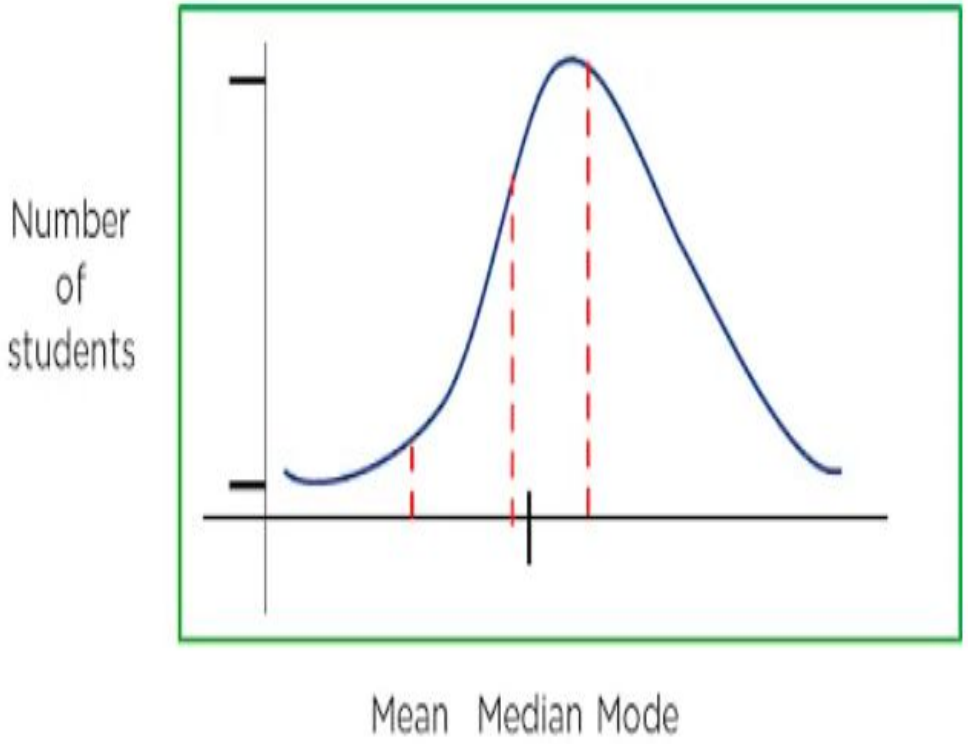


Figure: Negatively Skewed

Computation of Skewness

Pearson's coefficient

- The median is always the middle value, and the mean and mode are the extremes, so you can derive a formula to capture the horizontal distance between mean and mode.

- **Pearson's First Coefficient**: Pearson's first coefficient of skewness, or Pearson mode skewness, subtracts the mode from the mean and divides the difference by the standard deviation.

$$\text{Pearson's First Coefficient} = \frac{Mean - Mode}{Standard\ deviation}$$

- **Pearson's second coefficient** of skewness, or Pearson median skewness, subtracts the median from the mean, multiplies the difference by three, and divides the product by the standard deviation.

$$\text{Pearson's Second Coefficient} = \frac{3(Mean - Median)}{Standard\ deviation}$$

Computation of Skewness

1. If the Pearson's coefficient value is:
   - between -0.5 and 0.5 implies no skewness.
   - Data distribution is almost symmetrical
2. If the Pearson's coefficient value is:
   - between -1 and -0.5, the data is negatively and
   - 0.5 to 1, the data is positively skewed.
   - Data distribution is moderate skewness
3. If the Pearson's coefficient value is:
   - Lower than -1 (negatively skewed) and
   - Greater than 1 (positively skewed)
   - Data distribution is highly skewed.

Interpretation of the skewness

- Skewness is simply how much data set is deviating from its normal distribution.
- Larger negative value in the data set means that distribution is negatively skewed
- Larger positive value in the data set means that distribution is positively skewed.
- It is a good statistical measure that helps the investor to predict returns from the distribution.

Pearson’s coefficient

- Pearson’s First Coefficient: Pearson’s first coefficient of skewness, or Pearson mode skewness, subtracts the mode from the mean and divides the difference by the standard deviation.

$$\text{Pearson's First Coefficient} = \frac{Mean - Mode}{Standard\ deviation}$$

- The median is always the middle value, and the mean and mode are the extremes, so you can derive a formula to capture the horizontal distance between mean and mode.
- Pearson’s second coefficient of skewness, or Pearson median skewness, subtracts the median from the mean, multiplies the difference by three, and divides the product by the standard deviation.

$$\text{Pearson's Second Coefficient} = \frac{3(Mean - Median)}{Standard\ deviation}$$

If this value is between:

1. -0.5 and 0.5, the distribution of the value is almost symmetrical
2. -1 and -0.5, the data is negatively skewed, and if it is between 0.5 to 1, the data is positively skewed. The skewness is moderate.
3. If the skewness is lower than -1 (negatively skewed) or greater than 1 (positively skewed), the data is highly skewed.

Correlation

- Correlation is **a statistical term describing the degree to which two variables move in coordination with one another**.
- If the two variables move in the same direction, then those variables are said to have a positive correlation.
- If they move in opposite directions, then they have a negative correlation.
- In statistics, correlation or dependence is any statistical relationship, whether causal or not, between two random variables or bivariate data.

Scatter diagram and Linear Correlation

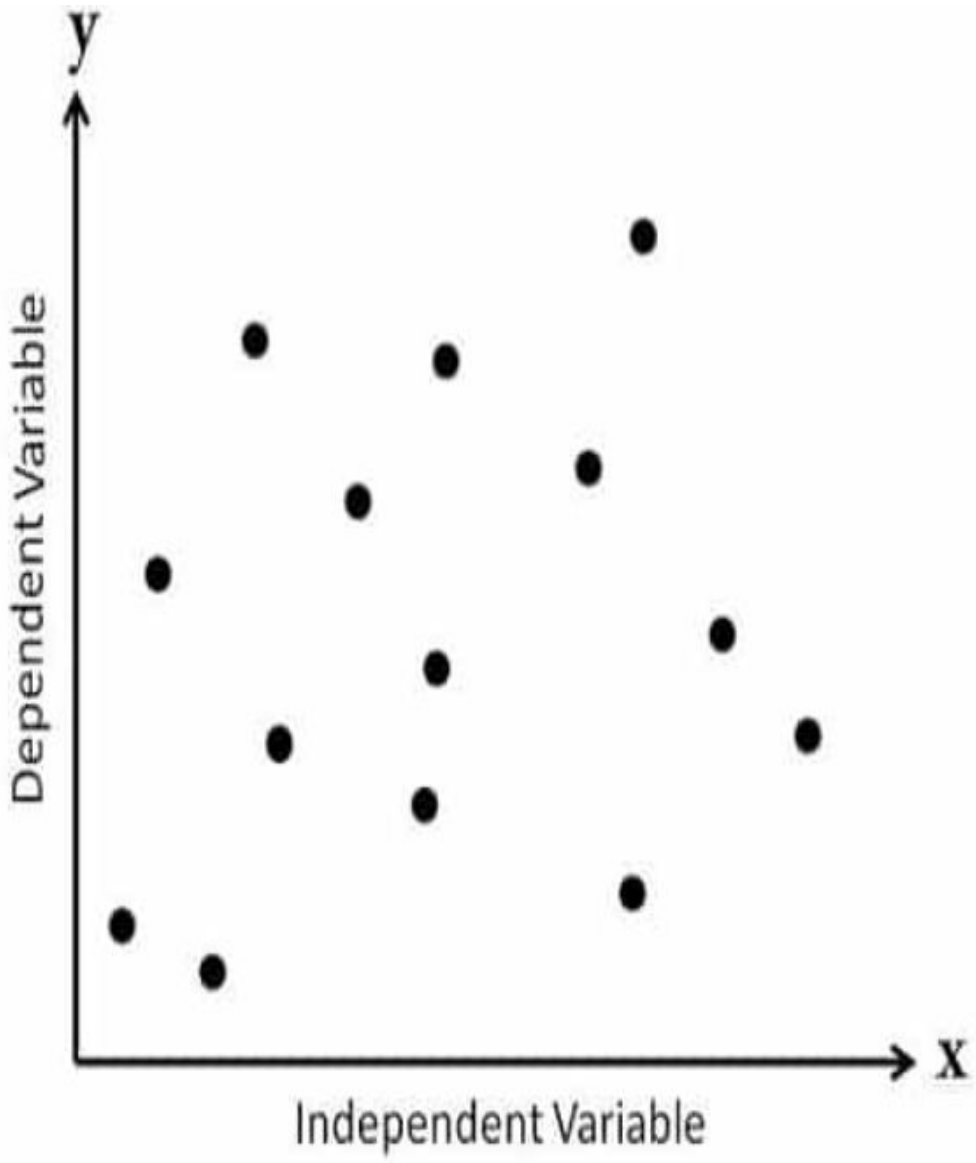
- In this topic, we will look at the following:
  - Explain what a scatter diagram is and how to make one
  - State what ‘strength’ and ‘direction mean with respect to correlations
  - Compute correlation coefficient r using the computational formula
  - Describe why correlation is not necessarily causation

Types of Correlation

- Positive, Negative or Zero Correlation
- Linear or Curvilinear Correlation:
- Scatter Diagram Method
- Pearson's Product Moment Co-efficient of Correlation
- Spearman's Rank Correlation Coefficient:

Scatter plot

- A scatter plot is **a tool for analyzing relationships between two variables for determining how closely the two variables are related**.
- One variable is plotted on the horizontal axis
- A second variable is plotted on the vertical axis
- The first variable is independent
- The second variable is dependent on the first.



Scatter diagram..cont'd

- Scatter diagrams are important in statistics because **they can show the extent of correlation, if any, between the values of observed quantities or phenomena** (called variables).

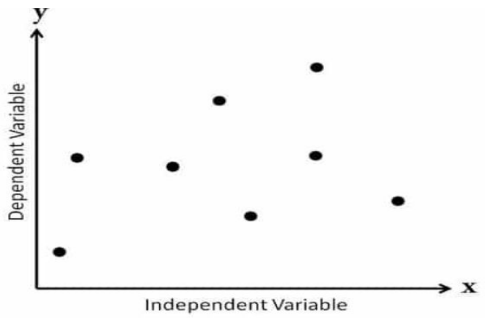


Figure 1: No correlation

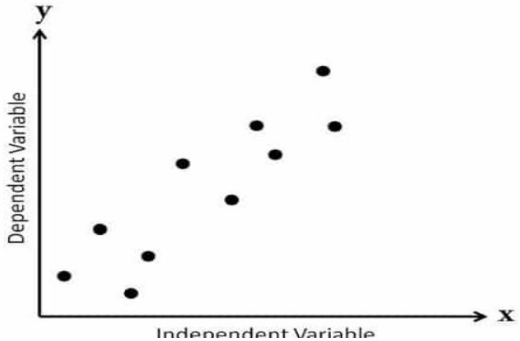


Figure 2: Moderate correlation

- According to the correlation, you can divide scatter diagrams into the following categories:
  - ❖ Scatter Diagram with No Correlation
  - ❖ Scatter Diagram with Moderate Correlation
  - ❖ Scatter Diagram with Strong Correlation

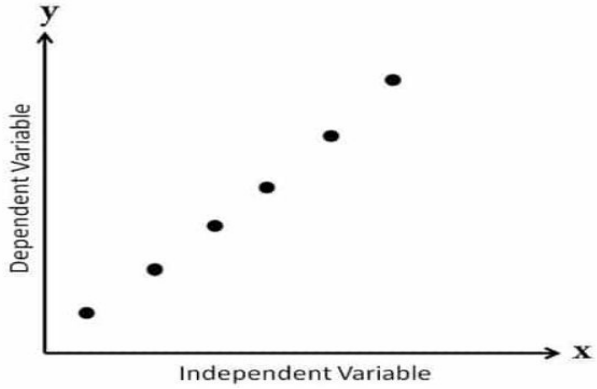
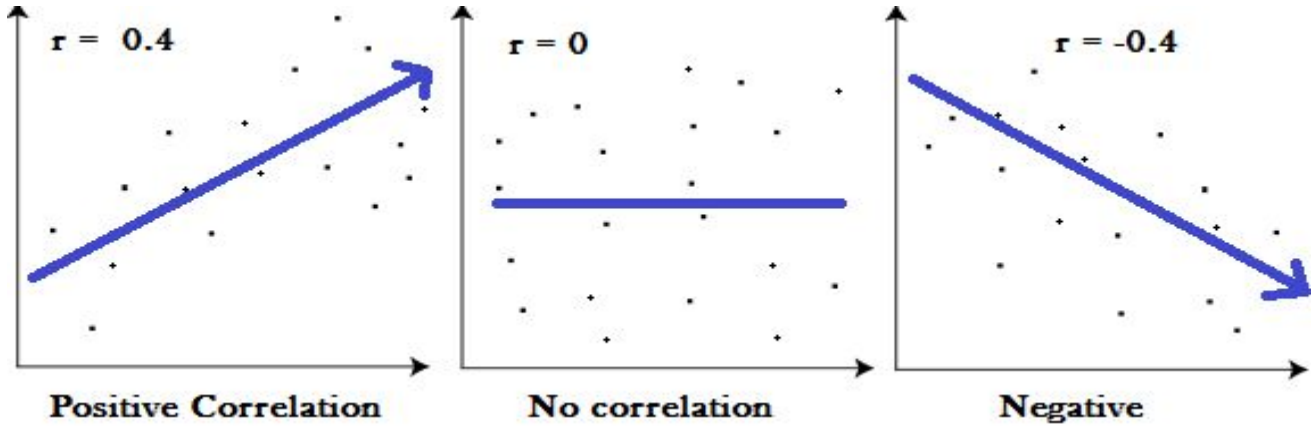


Figure 3: Strong correlation

Correlation

- Correlation analysis is used to quantify the degree to which two variables are related.



- A positive correlation is **a relationship between two variables in which both variables move in the same direction**.
- Therefore, when one variable increases as the other variable increases, or one variable decrease while the other decreases.
  - An example of positive correlation would be height and weight

Computation of correlation coefficient in Excel

- **Correlation coefficients** are used to measure how strong a relationship is between two variables

Using Excel to compute correlation

- **Step 1:** First you need to enable Data Analysis ToolPak in Excel.
- **Step 2:** Now click on **Data** followed by **Data Analysis**. A dialog box will appear.
- **Step 3:** In the dialog box select **Correlation** from the list of options.

- Step **4:** The Correlation menu will appear.

Computation of correlation

- **Step 5:** In this menu first provide the **Input Range**. The input range is the cell range of X and Y1 columns .
- **Step 6:** Also, supply the **Output Range** as the cell number where you want to display the result.
- **Step 7:** Check the **Labels** in first-row option if you have labels in the dataset. In our case column 1 has label X and column 2 has label Y1.
- **Step 8:** Click OK.

For example: Let's find the correlation coefficient for the variables and X and Y.

X	Y
1	10
2	4
3	5
4	7
5	8
7	6
8	5
9	9
Correlation coefficient=	0.02

Interpretation of correlation results

- How do you interpret correlation results?

  - **Degree of correlation:**

  - **Perfect correlation**:
    If the value is near ± 1, then it said to be a perfect correlation: as one variable increases, the other variable tends to also increase (if positive) or decrease (if negative).

  - **High degree correlation**
    If the coefficient value lies between ± 0.50 and ± 1, then it is said to be a strong correlation.

  - **Moderate degree correlation**
    If the coefficient value lies between ± 0.50 and ± 1, then it is said to be a strong correlation

Examples of Correlation Solutions

- Correlations identify a linear relationship between two variables.
  - What is the relationship between job satisfaction and job performance?
  - What is the relationship between hours studied and test grades?
  - What is the relationship between size of university and number of football team wins?

- Example: Computing the correlation is shown in Table below.

	Job Satisfaction	Self-Efficacy	Job Performance
Job Satisfaction	1		
Self-Efficacy	0.161551	1	
Job Performance	0.50922	0.205931	1

- **Case data set**
  - Job satisfaction and self-efficacy have a correlation of .16.
  - Job satisfaction and job performance have a correlation of .51.
  - Self-efficacy and job performance have a correlation of .21.

- However, how can we tell whether these are statistically significant?
  - **We obtain the P Value from Pearson (R) Calculator (**http://www.socscistatistics.com/pvalues/pearsondistribution.aspx)
  - The P-Value is .407055. The result is *not* significant at $p < .05$.

Two sample t-test

- The two-sample t-test (also known as the independent samples t-test) is a method used to test whether the unknown population means of two groups are equal or not.
- What are the 2 types of two sample t tests?
- An Independent Samples t-test compares the means for two groups. A Paired sample t-test compares means from the same group at different times (say, one year apart).
- How do you find a two sample t statistic?
- The test statistic for a two-sample independent t-test is calculated by taking the difference in the two sample means and dividing by either the pooled or unpooled estimated standard error. The estimated standard error is an aggregate measure of the amount of variation in both groups
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Two-Sample T-Test in Excel

- **How to find a p-value using the T.TEST function?**
- **Excel functions related to p-value**
- To use the function, you can type the formula:

- **=T.TEST(array 1, array 2, tails, type)**

- Where:

  - T.Test is the name of the function
  - Array 1 is the cell range of the first data set
  - Array 2 is the cell range of the second data set
  - Tails are the number of distribution tails
  - Type is the test that you want to be performed
- Case data

	Variable 1	Variable 2
Mean	88.5	75.3
Variance	55.83333	169.1222
Observations	10	10
Hypothesized Mean Difference	0	
Df	14	
t Stat	2.783079	
P(T<=t) one-tail	0.00733	
t Critical one-tail	1.76131	
P(T<=t) two-tail	0.01466	
t Critical two-tail	2.144787	

From results, we can identify that…

- The test statistic is: 2.783
- The p value is .015

Because our p-value is less than .05, we can reject the null and assume that a significant difference exists between our groups

ANOVA

One-Factor ANOVA in Excel

- An ANOVA is used to identify the mean difference between more than two groups, and a one-factor ANOVA is used to identify the mean difference between more than two groups when you have a single grouping variable and a continuous outcome.

- So, a one-factor ANOVA is used to answer questions that are similar to the following:
  - What is the mean difference in total output of five different industries?
  - What is the mean difference in performance of four different training groups?

- **Case data set**

ANOVA

- For example
  - effect size (F-value)= 117.391
  - p-value=, extremely small ($p < .0001$).
  - In the context of ANOVA implies significant difference among the means of our groups

- Group differences are large relative to the variances of the groups.
- The groups' variances each equal .767

Groups	Count	Sum	Average	Variance		
Column 1	10	19	1.9	0.766667		
Column 2	10	49	4.9	0.766667		
Column 3	10	79	7.9	0.766667		
ANOVA						
Source of Variation	SS	Df	MS	F	P-value	F crit
Between Groups	180	2	90	117.3913	4.8E-14	3.354131
Within Groups	20.7	27	0.766667			
Total	200.7	29				

- **Case data set**

Two-Way ANOVA in Excel

- An ANOVA is used to identify the mean difference between more than two groups
- A two-way ANOVA is used to identify the mean difference between more than two groups when you have a two grouping variables and a continuous outcome.

- The two-factor ANOVA is used to answer questions that are similar to the following:
  - What is the mean difference of test grades between left- and right-handed students, students in School A and School B, and the combinations of these groups?
  - What is the mean difference in total output of industries defined by location as well as industry?
  - What is the mean difference in performance for four different training programs, each performed at four different locations, and the combination of training program and location.

SUMMARY	Group 1	Group 2	Group 3	Total		
Group A						
Count	3	3	3	9		
Sum	27	22	24	73		
Average	9	7.333333	8	8.111111		
Variance	3	21.33333	3	7.361111		
Group B						
Count	3	3	3	9		
Sum	11	10	14	35		
Average	3.666667	3.333333	4.666667	3.888889		
Variance	21.33333	1.333333	16.33333	10.11111		
Group C						
Count	3	3	3	9		
Sum	9	17	30	56		
Average	3	5.666667	10	6.222222		
Variance	3	1.333333	0	10.44444		
Total						
Count	9	9	9			
Sum	47	49	68			
Average	5.222222	5.444444	7.555556			
Variance	14.94444	9.027778	10.27778			

First, we are going to determine whether the variable that defined our rows has a statistically significant effect. In the example dataset, this was Variable 1 (Group A, B, and C). To determine whether it was statistically significant, let's first look at the F-statistic, which is our test statistic.

(i) The F-statistic for Variable 1 was 5.127. While we would want to report it, the F-statistic doesn't tell us a lot on its own. So, we should look at the p-value.

The p-value was .017. Because this is less than .05, we consider the effect of Variable 1 to be statistically significant. So, there is a significant difference among Groups A, B, and C.

(ii) The F-statistic is 1.901. Again, we should report this value, but also look at the p-value to determine how to interpret our results.

The p-value is .178. Because this is greater than .05, our result is not statistically significant. Therefore, we would say that Variable 2 does not have a significant effect, and there is not a notable difference between Groups 1, 2, and 3.

(iii) Finally, The F-statistic is 1.660. Report this, but let's also look at the p-value.

The p-value is .203. This result is not statistically significant, and we would therefore say that there is no significant interaction between Variable 1 and Variable 2.

ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Sample	80.51852	2	40.25926	5.127358	0.017284	3.554557
Columns	29.85185	2	14.92593	1.900943	0.17824	3.554557
Interaction	52.14815	4	13.03704	1.660377	0.202861	2.927744
Within	141.3333	18	7.851852			
Total	303.8519	26				